

Autonomic Control of Heart Rate during and after Exercise

Measurements and Implications for Monitoring Training Status

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Abstract

Endurance training decreases resting and submaximal heart rate, while maximum heart rate may decrease slightly or remain unchanged after training. The effect of endurance training on various indices of heart rate variability remains inconclusive. This may be due to the use of inconsistent analysis methodologies and different training programmes that make it difficult to compare the results of various studies and thus reach a consensus on the specific training effects on heart rate variability. Heart rate recovery after exercise involves a coordinated interaction of parasympathetic re-activation and sympathetic withdrawal. It has been shown that a delayed heart rate recovery is a strong predictor of mortality. Conversely, endurance-trained athletes have an accelerated heart rate recovery after exercise. Since the autonomic nervous system is interlinked with many other physiological systems, the responsiveness of the autonomic nervous system in maintaining homeostasis may provide useful information about the functional adaptations of the body. This review investigates the potential of using heart rate

recovery as a measure of training-induced disturbances in autonomic control, which may provide useful information for training prescription.

Historically, the prescription of exercise training has largely been instinctive, with training volume, intensity and frequency being increased in a somewhat random manner in the belief that an increase in training will result in a concomitant improvement in performance and physical well-being. However, this vague approach to training prescription is not conducive to optimal physiological adaptation and performance improvement and may increase the likelihood of injury and symptoms of overtraining.^[1,2] Scientific research in this area may therefore contribute to the development of optimal training programmes.

Amongst other factors, the frequency, duration and intensity of exercise contribute to the nature and magnitude of the training effect, yet relatively little research has been conducted to quantify this effect. This is surprising since optimal training and rest regimens^[3] may only be planned once there is an understanding of the quantifiable effects of training. Physiological adaptations that occur in response to prolonged exercise training have been well documented in the literature. Some of these adaptations have been investigated as possible markers with which to monitor fitness, fatigue, overtraining and recovery. These include serum iron and ferritin concentrations,^[4-10] circulating testosterone and cortisol,^[2,11-13] intramuscular enzymes involved in fatty acid and glycolytic metabolism,^[14-22] and lactate metabolism^[23-26] to name a few. For more detail the reader is referred to the above-mentioned articles as this is beyond the scope of this article. It may be concluded, however, that many conflicting opinions exist about the effects of training and overtraining on each of the many physiological 'markers' that have been investigated. As such, no single measure has been identified that accurately assesses how an athlete is adapting and responding to a training programme.

Physical training is interpreted by the body as a biological stimulus.^[27] Accordingly, the physiological systems in the body, in particular the autonomic

nervous system and adrenal glands,^[28] respond during and after an exercise bout to maintain homeostasis. Repeated exposure to this training stress causes adaptations that are usually associated with improved performance. Thus, since the autonomic nervous system is interlinked with many other physiological systems, the responsiveness of the autonomic nervous system may provide useful information about the functional adaptations of the body.^[14,29] It is well known that the autonomic nervous system has a major effect on heart rate and the development of heart rate monitors^[30] has allowed further research into the mechanisms behind heart rate responses to exercise and adaptations to training. Particular attention has been focused on heart rate variability, largely as a means with which to evaluate cardiac autonomic control.^[31-36] Heart rate recovery is known to change with endurance training^[37-43] and in diseased states.^[44-50] However, there are no data to indicate whether heart rate recovery is a sensitive measure of autonomic control and in particular whether heart rate recovery can be used as a marker representing the body's capacity to respond to training.

This review describes the autonomic nervous system regulation of heart rate at rest and during exercise and particularly how this regulation changes with endurance training. We assess research that has investigated training-induced changes in heart rate variability and propose the use of heart rate recovery as a practical way to measure and monitor the training effect. A PubMed search of the academic literature was performed using the following terms: 'exercise heart rate recovery', 'autonomic control of heart rate exercise', 'endurance training adaptations', 'overtraining', 'heart rate and endurance training' and 'heart rate variability and exercise', limited to English papers and human subjects. The searches retrieved 3335 papers, of which 293 abstracts were reviewed and 142 full articles were evaluated.

1. Measurement of Autonomic Control of Heart Rate

1.1 Pharmacological Blockade

Pharmacological blockade has been used in research to assess the contributions of the parasympathetic and sympathetic nervous systems to the regulation of heart rate. The β -adrenergic blocking agents propranolol and metoprolol inhibit adrenergic stimulation of the heart via sympathetic nerves and circulatory catecholamines.^[32,51,52] Thus, the changes in heart rate observed after the administration of propranolol or metoprolol in healthy individuals may be attributed to the influence of an intact parasympathetic system and intrinsic heart rate. In the diseased heart, however, there may be other variables that make this interpretation more complex. Parasympathetic blockade is induced by the administration of atropine.^[32,33,52] The change in heart rate observed after the administration of atropine may be attributed to the influence of the sympathetic system and intrinsic heart rate. Conversely, the parasympathetic effect on heart rate may be investigated by assessing the difference in heart rate with and without atropine. Injections of propranolol and atropine together produce complete autonomic blockade,^[32,51] with which intrinsic heart rate can be assessed.^[52] Whilst these studies are important in identifying mechanisms of heart rate regulation, any performance data measured after pharmacological intervention should be interpreted with caution as the drugs may also have covert peripheral effects that may impact on performance.

1.2 Heart Rate Variability

Heart rate variability is a non-invasive measurement of the variation in the R-R intervals on an ECG. Interpretation of heart rate variability has also been recognized as a means with which to evaluate the autonomic control of heart rate.^[29] The indices of heart rate variability can be represented in a time domain in which R-R intervals (in milliseconds) are plotted against time (in seconds). Heart rate variability can also be represented in a frequency domain, which is the frequency at which the length of the R-R interval changes.^[31] The frequency domain is the

power spectral density and corresponds to the power of different oscillations present in the ECG. In the frequency domain, peaks at different frequencies represent differing contributions by the parasympathetic and sympathetic nervous systems. The parasympathetic activity contributes substantially to high frequency power (HF) [ms²], the energy in the heart rate power spectrum between 0.15 and 0.40 Hz. Both sympathetic and parasympathetic activity contribute to low frequency power (LF), 0.04–0.15 Hz.^[53] Ultra-low frequency power (ULF) between 0.00066 and 0.34 Hz) and very low-frequency power (VLF) between 0.000015 and 0.0033 Hz can also be measured.^[54] The ratio LF:HF represents sympatho-vagal balance, allowing heart rate variability to be used as an index of autonomic responsiveness, with high values reflecting sympathetic predominance.^[31]

2. Autonomic Regulation of Heart Rate: Transient Changes in Heart Rate Following the Start of Exercise

Pharmacological blockade and heart rate variability have been used to assess the parasympathetic and sympathetic contributions to the changes that occur in heart rate from rest through increasing intensities of exercise in healthy participants.^[31–36] For example, Polanczyk et al.^[32] investigated the contributions of sympathetic and parasympathetic modulation on heart rate variability during rest and exercise with the use of propranolol and atropine blockade.^[32] They concluded that time and frequency domain indices of heart rate variability were able to detect vagal activity, but not sympathetic activity at rest.^[32] However, other studies have shown that parasympathetic tone appears to predominate at rest^[32] and that the initial increase in heart rate at the onset of exercise is primarily due to parasympathetic withdrawal.^[34–36] As the intensity of exercise increases to more vigorous levels, the sympathetic system is activated such that at maximal intensity it is the combination of parasympathetic withdrawal and sympathetic activation that contribute to the observed tachycardia.^[34–36] Kannankeril et al.^[33] discovered that total parasympathetic withdrawal did not occur even during high-intensity exercise.^[33]

3. Chronic Response to Endurance Training: Changes in Heart Rate Regulation

3.1 At Rest

Many studies have shown that resting heart rate decreases slightly after endurance training.^[43,54-59] The mechanism behind the observed resting bradycardia in trained athletes was investigated by Smith et al.^[52] They suggested that this phenomenon was due to a decrease in the intrinsic rhythmicity of the heart and an increase in the predominance of parasympathetic control. There was also a slight decrease in the sympathetic contribution to heart rate.^[52] However, Melanson and Freedson^[60] found no significant difference in the resting heart rate of previously sedentary individuals after 16 weeks of training.^[60] A possible explanation for this is that the training load in this study may have been less than in studies that did observe slight decreases in resting heart rate. The above studies ranged in duration from 6 weeks to 5 months, the frequency of exercise bouts ranged from three to seven times per week and duration of each exercise bout ranged from 30 minutes to 4 hours.^[43,54-57,59] In the study by Melanson and Freedson,^[60] participants trained three times a week for 30 minutes per session and only increased the intensity of exercise from 70% heart rate reserve (HRR) in the first 2 weeks to 80% HRR from week 3. In the other studies, either the duration of each exercise session was increased substantially (high-intensity training at 70–90% maximum oxygen uptake [$\dot{V}O_{2\max}$] was increased by 130% and low-intensity training at <70% $\dot{V}O_{2\max}$ was increased by 100%^[55]) or the intensity was increased (from 50–60% HRR to 80–85% HRR^[57]) or both were increased (30 minutes of exercise at 55% $\dot{V}O_{2\max}$ in the first 2 weeks was increased to 50 minutes of exercise at 75% $\dot{V}O_{2\max}$ ^[56,59]).

Many studies reporting a difference in resting heart rate between trained and untrained individuals are cross-sectional in nature, thus limiting the ability to discern whether the difference is due to training or due to an inherent variation in the two populations.^[52,59] In addition, since changes in resting heart rate may be influenced by many environmental factors as well, the use of resting heart rate as a marker

of endurance training may be of limited value.^[61] Waldeck and Lambert^[62] investigated the repeatability of monitoring heart rate during sleep, since it can be assumed that the relationship between heart rate and training status should be more evident when the influence of extraneous factors is reduced.^[62] Sleeping heart rate was assessed over a period in which the subjects kept their daily training constant in order to determine if the measurement was precise enough to be used as a marker of training status. The intra-subject day-to-day variation of minimum heart rate during sleeping was about 5 beats/min. The authors concluded that this variation needs to be considered when interpreting training-induced changes in heart rate.^[62]

3.2 During Submaximal Exercise

It is widely accepted that, under controlled conditions, heart rate at the same submaximal intensity decreases after endurance training and is proposed to be due to a decrease in sympathetic activity to the heart.^[3,13,37,56,63-65] However, Swaine et al.^[66] found that a nonsignificant decrease in heart rate at a respiratory exchange ratio (R) of unity (heart rate/R = 1) was reversed after 8 weeks of moderate-intensity training suggesting that training-induced bradycardia occurs during the early stages of training and is subsequently reversed.^[66] Uusitalo et al.^[55] also detected nonsignificant changes in intrinsic heart rate and autonomic modulation (measured as the sympathovagal balance index after pharmacological blockade with atropine and propranolol) in female endurance athletes after 6–9 weeks of intense training.^[55] Swaine et al.^[66] proposed that training elicits different effects on various indices of fitness and that a decrease in submaximal heart rate may be a weak marker of cardiorespiratory fitness.^[66]

3.3 During Maximal Exercise

Some studies report that maximal heart rate ($HR_{\max}$) changes very little^[14,67-70] with endurance training, whereas other studies show decreases in $HR_{\max}$ with training^[71-73] and increases with detraining.^[74,75] However, a comprehensive review of studies that investigated the effects of endurance training on $HR_{\max}$, involving a total of 314 subjects, concluded that $HR_{\max}$ can decrease by between 5

and 13 beats/min with aerobic training, and increase by 4–10 beats/min with tapering or detraining in some individuals.^[76] Proposed mechanisms explaining decreases in HR_{max} with training include plasma volume expansion,^[72] enhanced baroreflex function^[77] and decreased β -adrenergic receptor number and density.^[78,79] However, further research is required to explain this phenomenon fully.^[76]

4. Heart Rate Variability

4.1 Effects of Endurance Training

Some cross-sectional studies using heart rate variability in the time domain have found that trained individuals have increased variability in R-R intervals than their untrained counterparts.^[54] However, in longitudinal studies or in cross-sectional studies using the frequency domain, differences in total power (TP), HF and LF due to training seem less conclusive. Some studies found that indices in the time domain^[54,60,80] or HF^[60,80] and LF^[81,82] in the frequency domain increased after training, where others did not see a change^[63,83] or reported a decrease in LF.^[80] These inconclusive results may be explained by the use of inconsistent methodologies such as the duration of the heart rate variability recording (6 minutes to 24 hours) and body positioning during recording (supine/semi-supine, during daily activities or sleep). Or differences in training such as the duration of the programme (6 weeks to 1 year), exercise session frequency (3–7 days per week), intensity (70–90% of HRR or HR_{max} that either remained constant or increased over the duration of the trial) and duration (25 minutes to 4 hours).^[54,57,60,80–84] This diversity in methodologies makes it difficult to compare results and come to a consensus regarding the effects of training on the various indices of heart rate variability (table I).

Bonaduce et al.^[54] compared 24-hour heart rate variability in detrained athletes versus untrained individuals, and also studied the effect of 5 months of vigorous training on heart rate variability in the same detrained athletes. Variability was higher in both the time and frequency domains, and LF : HF was lower in detrained compared with untrained participants, whereas the same variables were unchanged after 5 months of training in the detrained

athletes. There was, however, a decrease in resting heart rate after 5 months of training, suggesting that the change in resting heart rate after training may be due to mechanisms other than changes in sympatho-vagal balance. The authors suggest that, since the high frequency component represents “fluctuations in autonomic inputs to the heart rather than the mean level of autonomic inputs”,^[54] the possibility that training had increased parasympathetic influence without changing the modulation of efferent vagal activity cannot be excluded.^[54]

A similar phenomenon was observed in a study by Buchheit et al.,^[85] which investigated heart rate variability during slow-wave sleep in sedentary, moderately trained and highly trained young subjects. They found that moderately trained subjects had increased vagal-related heart rate variability indices compared with sedentary individuals, but that this increase was not perpetuated in highly trained athletes, despite the latter group having lower heart rate.^[85] Studies comparing habitually active elderly subjects to their sedentary counterparts revealed that those with a long-term ‘sportive’ lifestyle had higher global and vagal-related indices of heart rate variability.^[86,87] In addition, greater improvements in vagal control were gained when more time was spent in moderate- to high-intensity activities or when daily energy expenditure was increased to a moderate level of 600 kcal/day.^[87] These results may contribute to an improved understanding of the direct effect of training duration, frequency and intensity on heart rate variability. However, further investigation is required to confirm the findings and to assess their application when considering the use of heart rate variability to monitor training status.

4.2 Limitations

Besides the disparate findings of various studies due to the use of vastly different methodologies (table I), a number of other factors may influence the interpretation of heart rate variability and thus limit its use in monitoring training status. It has been found that heart rate variability at rest^[88] and 24-hour heart rate variability^[89] decreases with age and that there is less withdrawal of parasympathetic tone during submaximal and maximal exercise^[57] in the elderly. However, Buchheit et al.^[86] found that regular, long-term participation in sporting activities

Table I. Research studies investigating the effect of training on heart rate variability, showing the disparity in findings due to variation in the methodology of recording heart rate variability and in the training programmes prescribe

HR variability		frequency domain		Duration of measurement		Activity during measurement	Breathing control (breaths/min)	Duration of training	Exercise frequency	Exercise bout	
time domain	HF	LF	LF : HF	LF	LF : HF					intensity	duration
Cross-sectional studies											
Bonaduce et al. ^{[54] a}											
↑	↑	↑	↓	24 h	Daily activity and sleep	NR	1 mo detraining	5 ×/wk	NR	1 h	
Longitudinal studies											
Bonaduce et al. ^{[54] b}											
↔	↔	↔	↔	24 h	Daily activity and sleep	NR	5 mo	5 ×/wk	NR	4 h	
Levy et al. ^{[57] b}											
↑	↑ (absolute spectral value)	NR	NR	At rest = 30 min, during ex = last 2 min of each stage)	Supine rest and incremental exercise	NR	6 mo	4–5 ×/wk	↑ from 50–60% HRR to 80–85% HRR	45 min	
Loimaala et al. ^{[83] b}											
↔ (SDNN and pNN50)	↔	↔	NR	24 h	Daily activity and sleep		5 mo	Control = max 2 ×/wk, ex groups 1 and 2 = 4–6 ×/wk	Ex group 1 = 55% $\dot{V}O_{2max}$, ex group 2 = 75% $\dot{V}O_{2max}$	Minimum of 30 min at prescribed HR	
Carter et al. ^{[84] b}											
↑ (SDNN)	↑	↓	↑	Supine = 10 min, ex = 7 min	Supine rest and submaximal exercise	15	12 wk (2 × 4 wk training, + 2-wk taper)	4 ×/wk	70–90% HR _{max}	45–60 min (2nd 4-wk training ↑ by 20%)	
Melanson and Freedson ^{[60] b}											
↑ by wk 12 (pNN50 and rMSSD)	↑ by wk 12	↔	NR	10 min	Supine	10	16 wk	3 ×/wk	70% HRR in 1st 2 wk, 80% HRR for rest of study	30 min	
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Table I. Contd

HR variability		frequency domain		Duration of measurement	Activity during measurement	Breathing control (breaths/min)	Duration of training	Exercise frequency	Exercise bout		
time domain		HF	LF						LF : HF	intensity	duration
al Ani et al.[82] b											
↑ in 9 of 11 subjects, ↓ in 2 (the difference between maximum and minimum R-R intervals in 1 respiratory cycle)	Absolute spectral value ↑ in 9 of 11 subjects, ↓ in 2	NR	10 min	Semi-supine	Controlled at own frequency	6 wk	Daily	Intensity increased to match HR to 85% HR _{max}	25 min		
Tulppo et al.[80] c											
↑ (mean SDNN)	↑ in mean of normalized HF and absolute spectral value	↓ in mean of normalized LF, ↑ in absolute spectral value	24 h	Daily activity and sleep	NR	8 wk	6 ×/wk	70–80% HR _{max}	MI = 30 min, HI = 60 min		
Iwasaki et al.[81] d											
SDNN ↑ at 3 and 6 mo, ↓ at 9 and 12 mo	↔	↑ at 3 and 6 mo, ↓ at 9 and 12 mo	6 min	Supine	12	1 y	↑ over time	75–85% HR _{max}	↑ from 30–45 min 3–4 ×/wk to 7–9 h/wk		

a Detained vs untrained, normalization not reported.

b Normalization not reported.

c Normalized by the average R-R interval (CCV%).

d HFP and LFP normalized by dividing by total spectral power.

CCV = coefficient of component variance; **ex** = exercise; **HF** = high-frequency power; **HI** = high-intensity group; **HR** = heart rate; **HR_{max}** = maximal heart rate; **HRR** = heart rate reserve; **LF** = low-frequency power; **max** = maximum; **MI** = moderate-intensity group; **NR** = not reported; **pNN50** = percentage of adjacent R-R intervals that differ by more than 50 msec; **rMSSD** = root mean square of successive differences in R-R intervals; **SDNN** = standard deviation of normal R-R intervals; **VO_{2max}** = maximum oxygen uptake; ↓ indicates decrease; ↑ indicates increase; ↔ indicates no change.

may counteract the age-related decline in cardiac autonomic control.^[86] Thus, age, training status and their combined effects need to be considered when interpreting changes in the indices of heart rate variability. Carter et al.^[84] found that younger subjects had a larger autonomic adjustment to training than older participants of similar initial training status, suggesting that age may influence the adaptability of the cardiovascular system.^[84] Conversely, Levy et al.^[57] found that their older subjects, who had a 47% lower heart rate variability than the younger subjects before training, experienced a greater increase in heart rate variability (+68%) than the younger subjects (+17%) after 24 weeks of aerobic training.^[57] Tulppo et al.^[88] conducted a cross-sectional study assessing the effects of age and physical fitness on heart rate variability at rest and during exercise. They found that aging resulted in a reduction in heart rate variability at rest, but there were less profound age-related differences in heart rate variability during exercise. In addition, although there were no differences in heart rate variability at rest among three levels of fitness (based on $\dot{V}O_{2\max}$), the subjects with higher $\dot{V}O_{2\max}$ values had higher heart rate variability during exercise.^[88]

The dissociation between changes in heart rate and changes in vagal-related heart rate variability with training reported by Bonaduce et al.^[54] and Buchheit et al.^[85] may also limit the use of heart rate variability as a marker of cardiac vagal control when vagal activity is already high (in highly trained athletes).^[85] Buchheit et al.^[85] suggest further research is required to establish the mechanism behind the regression of vagal-related heart rate variability indices in highly trained athletes to levels comparable with sedentary individuals. Respiration may also affect heart rate variability as heart rate increases during inspiration and decreases during expiration.^[82] Therefore, respiratory frequency, for example, needs to be controlled to ensure that spectral sympathetic indices are unaltered.^[29] Tidal volume, environmental and body temperature,^[84] time of day,^[31] sampling frequency and method of normalization^[80] may all influence heart rate variability. Heart rate variability needs to be measured during controlled rest or exercise^[29,80] as the spectral analysis of heart rate requires steady-state conditions.^[29] Thus, in order for the results of various

studies to be compared and consensus reached regarding the specific training effects on heart rate variability, standardized methodologies need to be established for short- and long-term training involving untrained and trained individuals along with the relevant analysis.^[31,80,82]

5. Heart Rate Recovery After Exercise

Heart rate recovery is the rate at which heart rate decreases (or the time taken for heart rate to recover) after moderate to heavy exercise and is dependent on the relationship between parasympathetic and sympathetic nervous activity.

5.1 Mechanisms

The parasympathetic and sympathetic control of heart rate deceleration following the cessation of cardiovascular exercise has been investigated. Savin et al.^[90] proposed that sympathetic withdrawal contributed more to the decrease in heart rate immediately after exercise, with parasympathetic re-activation playing a greater role later in the recovery of heart rate.^[90] However, more recent studies have observed a coordinated interaction of parasympathetic re-activation and sympathetic withdrawal, with parasympathetic re-activation occurring faster, and therefore playing the more important role in the early deceleration of heart rate.^[33,39,49,91] The parasympathetic effect on heart rate recovery (measured as the difference in heart rate with and without atropine) was also investigated by Kannankeril et al.^[33,49] who found that parasympathetic reactivation occurred rapidly in the first minute of recovery.^[33] Parasympathetic stimulation then increased until 4 minutes and remained fairly constant until 10 minutes.

Heart rate recovery after maximal exercise is slower than after submaximal exercise in healthy individuals, and is attributed to the sympathetic nervous system being stimulated significantly more during maximal exercise. Thus, after high-intensity exercise, it is also the withdrawal of sympathetic stimulation that contributes to the deceleration of heart rate after the cessation of exercise.^[33] Pierpont et al.^[92] have investigated the time-course of autonomic change during immediate recovery from exercise and in particular the validity of using a first-

order exponential decay curve to model the heart rate response after exercise.^[92] They found this curve to be an imprecise depiction of heart rate recovery after maximal exercise, but acceptable for modelling recovery after submaximal exercise.^[92] They suggest that after low levels of exercise, when the sympathetic nervous system has not been significantly stimulated, heart rate recovery, which is governed mainly by the re-activation of the parasympathetic system, may follow a first-order exponential decay. However, after high levels of exercise, during which sympathetic stimulation dominates, the sympathetic drive may continue well into the recovery phase and contribute to sustained tachycardia despite the re-activation of the parasympathetic system. This would alter or delay the first order exponential decay that can be applied to heart rate recovery after submaximal exercise.^[92] When comparing heart rate recovery in boys and girls, Mahon et al.^[93] noted gender differences after submaximal exercise performed at an absolute work rate. However, after exercise that was performed at a relative intensity, heart rate recovery responses were similar. It is thus important to control the intensity of exercise that is performed directly before the assessment of heart rate recovery in order to more accurately evaluate changes in heart rate recovery within or between subjects.^[93]

5.2 Relationship to All-Cause Mortality

A delayed heart rate recovery has been found to be a strong predictor of subsequent mortality,^[44,45] especially where coronary heart disease and cardiovascular disease is concerned.^[46,47] Delayed heart rate recovery is defined in this context as a reduction of 12 beats or less in the first minute following symptom-limited maximal treadmill exercise using standard and modified Bruce protocol.^[45] Desai et al.^[94] found no correlation between abnormal heart rate recovery and increased incidence of myocardial infarction or ischaemia.^[94] However, they found strong correlations between heart rate recovery and chronotropic variables of peak heart rate, percentage peak heart rate and percentage HRR in both healthy individuals and groups with coronary artery disease.^[48] This suggests that heart rate recovery depends significantly on the chronotropic response, such that chronotropic incompetence will produce

an abnormally slow heart rate recovery.^[48] Gaibazzi et al.^[95] found no prognostic value in the use of 1-minute heart rate recovery after symptom-limited maximal exercise on a cycle ergometer;^[95] however, a retrospective study of 2193 men who were followed-up after 7 years, found that the first 2 minutes of heart rate recovery after treadmill testing did predict mortality.^[46] This study also found that there was an increased risk of coronary artery disease in the subjects who had a slower recovery of heart rate during the second minute of recovery after symptom-limited graded treadmill exercise to 85% of their estimated HR_{max}.^[46] One mechanism proposed to explain the relationship between delayed heart rate recovery (delayed parasympathetic re-activation) and mortality is that during delayed recovery the protective effect that parasympathetic tone affords the cardiovascular system is diminished.^[49] Kannankeril and Goldberger^[49] suggest that parasympathetic tone may produce an 'antiarrhythmic' effect by prolonging ventricular refractoriness, but this mechanism requires further investigation during exercise and recovery in healthy and diseased populations.^[49]

Seshadri et al.^[50] suggested that a delayed 1-minute heart rate recovery after exercise may be a useful prognostic tool for patients with chronic obstructive pulmonary disease, as obstructive, restrictive and mixed-type lung disease may be associated with autonomic imbalance. If this is the case, it may become possible to differentiate between all-cause mortality that is due to underlying pulmonary disease from mortality due to cardiovascular causes.^[50] Kizilbash et al.^[96] investigated the relationship between heart rate recovery and the metabolic syndrome and found that slower heart rate recovery does not precede development of the metabolic syndrome, but rather appears after syndrome components are present.^[96]

The effect of cardiac rehabilitation programmes on heart rate recovery has also been investigated. Tiukinhoy et al.^[97] and Giallauria et al.^[98-100] found that heart rate recovery improved after participation in cardiac rehabilitation programmes. They concluded that heart rate recovery, a simple means with which to assess autonomic tone, may be used to identify patients at higher risk for cardiac events, to evaluate patient outcomes after cardiac rehabilita-

tion programmes and to investigate possible mechanisms behind the beneficial prognostic effects of exercise training in cardiac patients.^[97-100] MacMillan et al.^[101] also found significant improvements in heart rate recovery after exercise training during cardiac rehabilitation and noted that heart rate recovery improved similarly in older and younger males and females.^[101] An interesting question that has not been resolved is whether exercise training alone improves mortality, in contrast to whether the improvement in heart rate recovery, as a result of an adjustment between sympathetic and parasympathetic control, has an independent role in improving mortality.

5.3 Relationship to Training Status

Endurance-trained athletes have faster heart rate responses at the start of exercise and after the cessation of exercise compared with sedentary subjects.^[37-39] As such, heart rate recovery may be useful as an indicator of the functional status of the cardiovascular system. In support of this assertion, Sugawara et al.^[40] found that post-exercise heart rate recovery was significantly accelerated after only 8 weeks of moderate-intensity training and that it returned to baseline levels after 4 weeks of subsequent detraining.^[40] Yamamoto et al.^[41] associated an increase in parasympathetic tone with the accelerated heart rate recovery they observed within just 7 days of a 6-week endurance training period.^[41] They found that the heart rate variability index SDNN (the standard deviation of normal R-R intervals) and the $\ln$ LF power (the natural logarithm of low frequency power), measured 10 minutes after the cessation of exercise, increased with endurance training.^[41] In this study, post-exercise heart rate recovery remained accelerated after the first week, whereas resting heart rate continued to decrease. They concluded that a training-induced change in the autonomic control of heart rate recovery occurs sooner than it does for resting heart rate.^[41] The effect that an early training-induced increase in plasma volume may have had on the observed adaptations to heart rate was not considered in this study.

Although most of the research investigating training-induced changes in heart rate recovery has focused on endurance training, Otsuki et al.^[102] found that heart rate recovery is accelerated in both

strength- and endurance-trained athletes.^[102] In their study, the time constant of post-exercise heart rate decay was lowest in endurance-trained athletes followed by strength-trained athletes, with the heart rate decay of both these groups being significantly lower than that of the controls.^[102] Brown and Brown^[103] studied the heart rate variability during recovery after high-intensity exercise in trained masters runners and showed that heart rate variability decreased in both time and frequency domains. These findings suggested an increased parasympathetic withdrawal during heart rate recovery after high-intensity exercise in trained masters runners.

5.4 Practical Applications

Whereas the analysis of heart rate during exercise may provide a measure of cardiac load, heart rate recovery may provide a quantifiable measure of disturbances in autonomic control in response to endurance training. Measuring heart rate recovery on a regular basis after a standardized submaximal test under controlled conditions may enable coaches to monitor how athletes are responding to their training, thereby facilitating optimal prescription of training. A submaximal shuttle running test (Heart rate Interval Monitoring System [HIMS]), which has been designed in our laboratory, comprises four 2-minute stages (each stage increasing in intensity relevant to the previous one) separated by recovery periods of 1 minute.^[104] The subjects are asked to run between two lines drawn 20 m apart on a rubberized indoor floor. The running pace of each of the four stages (8.4, 9.6, 10.8 and 12.0 km/h, respectively) is set by a pre-recorded auditory signal on a compact disk. Heart rate is recorded throughout the test and heart rate recovery during the first minute after the cessation of the fourth running stage is analysed. Although we are not aware of any published articles investigating the effect of extraneous factors on heart rate recovery, an effort is made to produce controlled conditions in which this test is administered. For example, subjects should refrain from ingesting caffeine or participating in strenuous exercise at least 2 hours before starting the submaximal test, and they should begin the test in a euhydrated state. During each recovery stage the athletes stand in an upright, stationary position with their hands by their sides and do not talk. We

acknowledge that this test may be best suited for individuals who are trained in running (particularly when monitoring highly trained athletes) and that a more sensitive assessment of changes in heart rate recovery over time might be attained if sport-specific submaximal tests of this kind were developed. The HIMS test was designed to be submaximal, easy and quick to complete and non-aversive for the athletes so that it can be administered regularly during different phases of training. The reliability of the test has been established, with the standard error of measurement being just over 1%.^[104] For the information to be useful, the test needs to be done on a weekly basis so that a heart rate profile can be established for each athlete. While the heart rate recovery test on its own may not be diagnostic, if the data are interpreted in context with other data gathered (i.e. training load, subjective symptoms) it will provide useful information and contribute to decisions about training prescription.

6. Conclusions

Despite years of research, no single factor has been identified that can quantify how an individual is responding to training or predict overtraining with accuracy. There is therefore no quantitative means with which to assess whether the training response is optimal in terms of producing advantageous physiological adaptations and improving performance. The usefulness of a physiological marker also depends on the ease and frequency with which it can be measured and on the speed with which the results can be interpreted so that frequent monitoring is possible with little inconvenience to the athlete. The correlation between training and the observed changes in these physiological variables is highly personal and dependant on the individual tolerance to an exercise load, which may be a culmination of many internal and external factors. Thus, attention needs to be directed towards the identification of markers that reflect an individual's overall responsiveness or adaptability to training rather than an absolute measure of the changes in specific physiological variables in response to exercise. The functional state of the autonomic nervous system may provide useful information about the overall functional adaptation of the body in response to a train-

ing stimulus as it is a system interlinked with many other physiological systems.

This review therefore began by examining the changes that occur in heart rate with endurance training and assessed the usefulness of heart rate based physiological measurements as tools with which to monitor training status. There is general consensus that resting and submaximal heart rates decrease with endurance training, although changes in HR_{max} are less clear. All of these measurements, however, are affected by extraneous factors that limit their use as sensitive markers of training status. More insight into the autonomic control of heart rate (in particular the contributions of the parasympathetic and sympathetic systems to heart rate) has become possible with investigations involving heart rate variability. The effect of endurance training on the various indices of heart rate variability, however, remains inconclusive. This can be attributed to the use of inconsistent methodologies that make it difficult to compare results between different studies and to reach consensus.

The autonomic control of heart rate recovery following the cessation of cardiovascular exercise has also been investigated. Since the rate of recovery is determined by the coordinated interaction of parasympathetic and sympathetic systems, heart rate recovery may provide a measure of the disturbance in autonomic control in response to endurance training. It has been found that heart rate responses both at the onset and cessation of exercise are faster in endurance-trained athletes compared with untrained individuals^[37,38] and that a delayed heart rate recovery is associated with mortality.^[44,45] Thus, the rate of heart rate recovery may represent a continuum of autonomic nervous system function with delayed heart rate recovery, compromised autonomic nervous system function and mortality at the one extreme and accelerated heart rate recovery, optimal autonomic function and enhanced cardiovascular condition on the other end of the continuum. Whereas the analysis of heart rate during exercise may provide a measure of cardiac load, the measurement of heart rate recovery may reflect training-induced disturbances in the autonomic nervous system that provide useful information about how an athlete is responding to endurance training. The rate of decay of heart rate during the first minute after the cessa-

tion of a standardized cardiovascular test may be unique to each individual and may be an easily measurable index of an athlete's current cumulative state of fatigue. It will provide useful information on cardiac and autonomic nervous system adaptations that may contribute to a more comprehensive understanding of how an individual is responding to training. This method of quantifying an individual's physiological responsiveness to training and its practical application in the prescription of optimal training programmes requires further investigation.

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